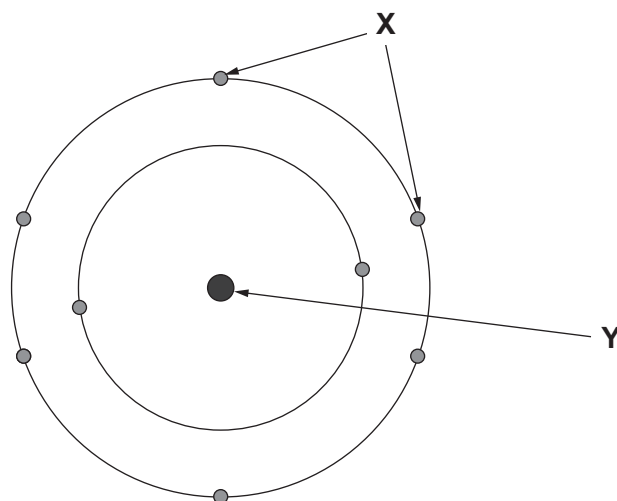


*Answer all questions.*

1. (a) The following diagram shows the structure of an atom.



|           |         |        |          |         |
|-----------|---------|--------|----------|---------|
| electrons | protons | shells | neutrons | nucleus |
|-----------|---------|--------|----------|---------|

Complete the following sentences using words from the box.

[3]

The particles labelled **X** are called .....

The part of the atom labelled **Y** is called the .....

It contains particles called ..... and .....

- (b) Draw a line from each particle to the charge of that particle.

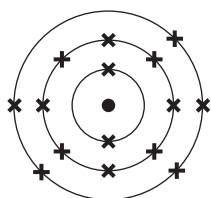
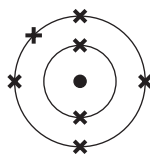
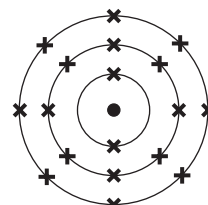
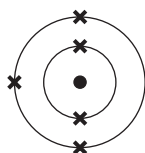
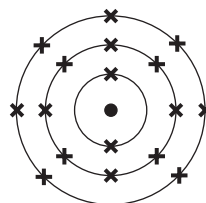
[2]

|          |    |
|----------|----|
| proton   | 0  |
| neutron  | +1 |
| electron | -1 |



(c) The diagrams below show five different atoms labelled **A-E**.

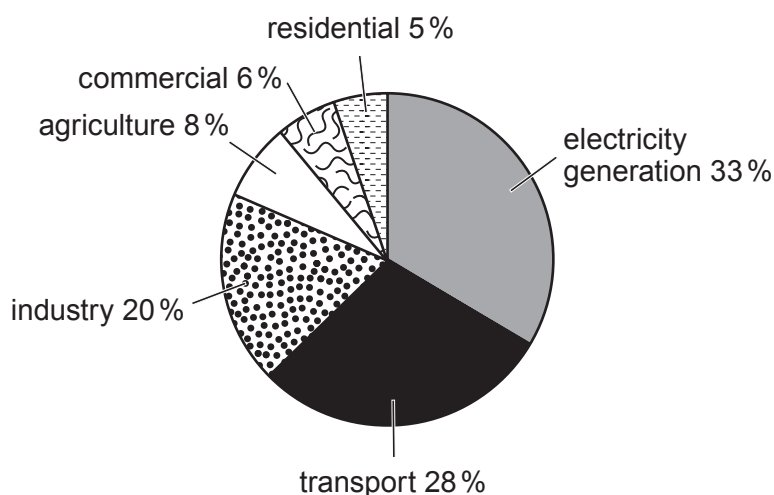
These letters are **not** the chemical symbols for the elements.

**A****B****C****D****E**

Use the letters **A-E** to complete the following sentences.

- (i) The atoms found in Group 5 are ..... and ..... [1]
- (ii) The atom with an atomic number of 7 is ..... [1]
- (iii) The atom that is an inert gas is ..... [1]
- (iv) The atoms found in Period 2 are ..... and ..... [1]





**Figure 3**

(a) Draw a trend line on **Figure 1**. [1]

(b) Suggest why there was a large rise in the amount of carbon dioxide in the atmosphere after around 1950. [2]

.....

.....

.....

.....

(c) Give **one** way in which we are trying to reduce carbon dioxide emissions. [1]

.....

.....

(d) Some people say that changes in carbon dioxide levels are **not** responsible for global warming. Tick (✓) the statement that supports this opinion. [1]

Most carbon dioxide is produced by electricity generation

☐

Between 1900 and 2010 there was a massive increase in industry

☐

The mean global temperature remained constant between 1950 and 1980

☐

The average number of cars per home has increased steadily since the 1980s

☐

The use of energy efficient appliances has increased since 2000

☐

|   |
|---|
|   |
| 5 |

